

Examination Preparation Guide
DSA 8505: Bayesian Statistics
Strathmore Institute of Mathematical Sciences

Key Preparation Guidelines

1. Master the Core Distributions

- Be able to extract mean, variance and standard deviation directly from:

$$N(\mu, \sigma^2), \quad \text{Gamma}(\alpha, \beta), \quad \text{Beta}(\alpha, \beta).$$

- For a Normal posterior, compute a 95% credible interval quickly:

$$\mu \pm 1.96\sigma.$$

- Understand that Gamma distributions are skewed.
- Know that HPD intervals are preferred for skewed posteriors.

2. Be Comfortable with Conjugate Priors

- Beta-Binomial update:

$$\text{Beta}(\alpha, \beta) \rightarrow \text{Beta}(\alpha + x, \beta + n - x).$$

- Normal-Normal posterior variance:

$$\sigma_n^2 = \left(\frac{1}{\sigma^2} + \frac{1}{\sigma_0^2} \right)^{-1}.$$

- Normal-Normal posterior mean:

$$\mu_n = \frac{\frac{x}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_0^2}}.$$

- Always update **both** prior parameters.

3. Understand Bayesian Estimators

Loss Function	Bayes Estimator
Squared Error	Posterior Mean
Absolute Error	Posterior Median

Do not confuse mean and median.

4. Know How to Compute and Interpret Bayes Factors

- Compute:

$$BF = \frac{P(D|H_1)}{P(D|H_0)}.$$

- Interpretation guide:
 - 1–3: Weak evidence
 - 3–10: Moderate evidence
 - >10: Strong evidence
- Always interpret results in words.

5. Be Fluent in Bayes' Theorem

- Write marginal probability clearly:

$$P(+) = P(+|D)P(D) + P(+|D^c)P(D^c).$$

- Compute posterior:

$$P(D|+) = \frac{P(+|D)P(D)}{P(+)}.$$

- Understand the base-rate effect in diagnostic problems.

6. Practice Completing the Square

- Derive the Normal–Normal posterior from scratch.
- Avoid mixing variance and standard deviation.
- Show algebra clearly.

7. Understand Regression Concepts

- OLS slope formula:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}.$$

- Matrix form:

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

- Be able to write the likelihood for regression.
- Understand why MCMC is required (non-closed form posterior, high-dimensional integration).

8. Bayesian Model Averaging (BMA)

$$\hat{\beta}_{BMA} = \sum_k P(M_k|D) \hat{\beta}_k.$$

BMA is a weighted average using posterior model probabilities.

9. Show Working Clearly

- Define symbols.
- Show intermediate steps.
- Box final answers.
- Interpret results briefly.

10. Avoid Common Mistakes

- Mixing credible and confidence intervals.
- Forgetting false positives in medical problems.

- Confusing variance and standard deviation.
- Arithmetic errors in Bayes' theorem.
- Writing formulas without explanation.

11. Time Management Strategy

- Start with shorter computational questions.
- Attempt derivations while fresh.
- Leave time for interpretation.

12. Revision Strategy

- Re-derive Normal-Normal posterior at least three times.
- Practice Beta-Binomial problems.
- Solve at least two diagnostic-test problems.
- Practice one regression example by hand.

*Bayesian statistics is about updating beliefs logically.
If you understand the logic of updating, the formulas become easy.*